

		Vertical resultant force = $10.0 - 25.0 \cos 45^\circ = 7.7 \text{ N}$ Resultant = $(12.3^2 + 7.7^2)^{1/2} = 14.5 \text{ m}$
2	D	Velocity is the gradient of the Displacement time graph.
3	D	Gradient of displacement time graph is velocity, hence option A, B and C are incorrect. Since displacement decreases after 2.5s, option D is true